TECHBIG SOLUTIONS

Ambattur, Chennai

Tableau – Complete E2E Training Syllabus

Data Visualization | Analytics | Tableau Desktop | Tableau Cloud/Server | Tableau Prep

Program Objective: Build complete practical Tableau capability from data connection and preparation through data modeling, calculated fields, LOD expressions, advanced analytics, interactive dashboards, Tableau Server/Cloud, security, governance, performance and production support.

Reference basis: Tableau learning and product documentation covering Tableau Desktop, Tableau Prep, data modeling, calculations, dashboards, Tableau Cloud/Server, governance, performance and analytics.

Program	Tableau – Complete E2E
Level	Beginner → Advanced | Data Analyst / BI Developer / Tableau Consultant
Core Areas	Desktop | Prep | Data Modeling | Calculations | Dashboards | Analytics
Advanced Areas	LOD | Table Calculations | Server/Cloud | Security | Governance | Performance | APIs
Suggested Duration	100–140 hours, customizable by batch and project depth

1. Tableau Overview & Business Intelligence

• Tableau ecosystem: Desktop, Server, Cloud, Public and Prep

• BI, analytics and self-service reporting concepts

• Workbooks, worksheets, dashboards and stories

• Dimensions, measures, discrete and continuous fields

• End-to-end analytics lifecycle

2. Tableau Desktop – Fundamentals

• Installation and interface

• Connecting to data

• Data pane and Analytics pane

• Marks card and shelves

• Rows, Columns, Pages and Filters

• Formatting, captions, tooltips and worksheet organization

3. Data Sources & Connectivity

• Excel, CSV, text and JSON sources

• SQL Server and relational databases

• Cloud databases and warehouses

• Web and published data sources

• Joins, relationships and unions

• Live vs Extract connections

• Credentials and connection management

4. Tableau Prep – Data Preparation

• Tableau Prep Builder overview

• Input, clean, aggregate and output steps

• Data profiling and quality

• Cleaning and standardization

• Joins, unions and pivots

• Calculated fields in Prep

• Flows, schedules and reusable data-preparation processes

5. Data Modeling in Tableau

• Relationships vs joins

• Logical and physical layers

• Cardinality and referential integrity

• Blending concepts

• Star and dimensional models

• Multi-table analytical models

• Model validation and performance considerations

6. Calculated Fields – Fundamentals

• Calculation syntax

• String, numeric and date calculations

• IF, CASE and logical expressions

• Null handling

• Type conversion

• Row-level calculations

• Reusable business logic

7. Advanced Tableau Calculations

• Aggregate vs row-level calculations

• Level of Detail (LOD) expressions

• FIXED, INCLUDE and EXCLUDE LODs

• Nested calculations

• Table calculations

• WINDOW, LOOKUP and INDEX functions

• Advanced conditional and analytical calculations

8. Filters & Filtering Strategy

• Dimension and measure filters

• Context filters

• Data-source filters

• Extract filters

• Table calculation filters

• Top N and conditional filters

• Filter order of operations

• Cascading and dependent filters

9. Charts & Visualizations

• Bar, line, area and combo charts

• Scatter plots

- Pie, treemap and packed bubbles
- Maps and geographic charts
- Heat maps and highlight tables
- Box plots and histograms
- Waterfall, funnel and Gantt charts

10. Advanced Visual Analytics

- Reference lines, bands and distributions
- Forecasting
- Trend lines and clustering
- Analytics pane
- Parameters and parameter actions
- Set actions
- Viz-in-tooltip and advanced interaction patterns

11. Dashboard Design & UX

- Dashboard layout and sizing
- Tiled vs floating objects
- Device-specific layouts
- Navigation and buttons
- Show/hide containers
- Actions and interactivity
- Dashboard performance and design standards

12. Tableau Stories

- Story points and narrative flow
- Storytelling with data
- Interactive business narratives
- Executive presentation design
- Annotations and insights
- Publishing stories

13. Tableau Maps & Geospatial Analytics

- Geographic roles
- Latitude and longitude
- Symbol and filled maps
- Map layers
- Spatial files and custom territories
- Density maps
- Location hierarchies and geographic filtering

14. Time-Series & Time-Based Analytics

- Date parts vs date values
- Date hierarchies
- YTD, MTD and QTD analysis

- Period-over-period comparisons
- Running totals and moving averages
- Fiscal calendars
- Trend and seasonality analysis

15. Tableau Parameters & Sets

- Parameter creation and data types
- Dynamic parameters
- Parameter-driven calculations
- Sets and set membership
- Combined sets
- Set actions
- Dynamic Top N and segmentation

16. Table Calculations

- Quick table calculations
- Addressing and partitioning
- Running total, difference and percent difference
- Rank and percent of total
- WINDOW functions
- LOOKUP and moving calculations
- Troubleshooting table calculations

17. Tableau Extracts & Hyper

- Extract fundamentals
- Hyper technology overview
- Full vs incremental extracts
- Extract filters
- Extract refresh schedules
- Extract optimization
- Extract encryption and management

18. Tableau Server & Tableau Cloud

- Server/Cloud architecture
- Sites, projects and workbooks
- Users, groups and permissions
- Publishing content
- Data sources and connections
- Subscriptions and alerts
- Content ownership and administration

19. Security & Governance

- User and group permissions
- Project and workbook security
- Row-level security patterns

• User filters and entitlement tables

• Data-source security

• Certified data sources

• Governance standards and audit concepts

20. Tableau Prep & Server Integration

• Publishing Prep flows

• Flow ownership and permissions

• Scheduling and automation

• Flow monitoring

• Output management

• Error handling and operational support

21. Performance Optimization

• Workbook performance concepts

• Performance Recorder

• Extract optimization

• Query reduction

• Calculation optimization

• Dashboard load optimization

• Filter and visualization optimization

• Server-side performance

22. Tableau APIs & Automation

• Tableau REST API overview

• Metadata API concepts

• Embedding and trusted access concepts

• Content automation

• User/group automation

• Workbook/data-source deployment

• Automation and integration scenarios

23. Tableau with SQL & Data Warehouses

• SQL fundamentals for Tableau

• Views and stored procedures

• Joins and aggregations

• CTEs and window functions

• Source-side calculations

• Cloud warehouse connectivity

• Query performance and pushdown strategies

24. Tableau with Salesforce & Enterprise Data

• Salesforce data connectivity

• CRM analytics scenarios

• ERP and enterprise-data integration

• Published data sources

• Centralized semantic/business data

• Cross-system KPI reporting

25. Tableau AI & Modern Analytics

• Tableau AI capabilities overview

• Natural-language analytics concepts

• AI-assisted insights and explanations

• Responsible AI

• Validation of AI-generated insights

• Governance for AI-enabled analytics

26. Administration & Monitoring

• Server/Cloud administration concepts

• Jobs and background tasks

• Refresh monitoring

• Usage and activity metrics

• Content monitoring

• License and user management

• Operational troubleshooting

27. Deployment & DevOps

• Development, test and production environments

• Workbook promotion

• Project/content migration

• Version control concepts

• Automation and CI/CD concepts

• Release management

• Backup and recovery considerations

28. Implementation Methodology

• Requirement gathering

• KPI and business-rule definition

• Source-system assessment

• Data-model design

• Prototype and dashboard design

• Testing and UAT

• Deployment and user training

• Go-live and hypercare

29. Real-Time E2E Projects

• Sales dashboard: SQL/Excel → model → calculations → executive dashboard

• Finance dashboard: revenue, cost, P&L;, budget vs actual and variance

• HR analytics: headcount, hiring, attrition and department KPIs

• Inventory analytics: stock, aging, movement and replenishment

• Procurement dashboard: spend, suppliers, PO and delivery performance

• Production dashboard: output, downtime, quality and efficiency

• Customer analytics: segmentation, retention and profitability

• Salesforce/CRM analytics implementation

• Enterprise Tableau Server/Cloud deployment with RLS and governance

• End-to-end production support and performance-optimization project

30. Troubleshooting & Production Support

• Data-connection failures

• Extract refresh failures

• Incorrect joins/relationships

• Calculation discrepancies

• Filter and LOD issues

• Slow dashboards

• Permission and row-level-security issues

• Publishing failures

• Server/Cloud refresh and job failures

• Root-cause analysis and incident management

31. Interview & Certification Preparation

• Tableau interview fundamentals

• Calculated-field and LOD questions

• Dashboard design case studies

• Performance-tuning scenarios

• Tableau Server/Cloud administration questions

• Security and governance scenarios

• Real-time project explanation

• Mock interview and practical assessment

• Tableau Certified Data Analyst preparation

Recommended Hands-on Deliverables

• Build an executive sales dashboard from Excel/SQL through data preparation, modeling, calculations and interactive visualization.

• Create advanced LOD and table-calculation solutions for business KPIs.

• Develop a Tableau Prep flow and publish/operate it in a production environment.

• Implement row-level security, permissions, published data sources and governed content.

• Optimize a slow workbook using Performance Recorder, extract design, calculations and dashboard techniques.

• Complete a development → test → production deployment with monitoring and production-support procedures.

Training Outcome

Learners completing this program should be able to connect and prepare data, build robust Tableau models and calculations, create professional dashboards and analytical stories, publish and secure content on Tableau Server/Cloud, optimize performance and support real-world BI implementations.